

Two-qubit gates and observable

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NOT
0 $\longleftrightarrow$ 1

X
 $|0\rangle \longleftrightarrow |1\rangle$

AND

a	b	$a \cdot b$
0	0	0
0	1	0
1	0	0
1	1	1

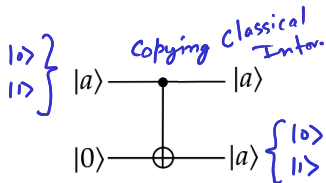
modulo-2 mul.

a	b	$a \oplus b$
0	0	0
0	1	1
1	0	1
1	1	0

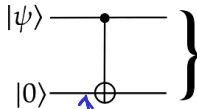
video

Theorem "No-cloning of an 'unknown' quantum state"

$$\langle \Psi | \Psi \rangle = 1 = |\alpha_0|^2 + |\alpha_1|^2 = 1$$



$$a \in \{0, 1\}$$



Entangled State
except $\alpha_0 = 0$
or $\alpha_1 = 0$

$$|\psi\rangle = \alpha_0 |0\rangle + \alpha_1 |1\rangle$$

$$|\Psi\rangle = \alpha_0 |00\rangle + \alpha_1 |11\rangle$$

$$|\psi\rangle|0\rangle = \alpha_0 |00\rangle + \alpha_1 |10\rangle$$

$$|\Psi\rangle \neq \underbrace{(|\psi\rangle \otimes |\psi\rangle)}$$

$$(\alpha_0 |0\rangle + \alpha_1 |1\rangle) \otimes (\alpha_0 |0\rangle + \alpha_1 |1\rangle) = \alpha_0^2 |00\rangle + \alpha_0 \alpha_1 (|01\rangle + |10\rangle) + \alpha_1^2 |11\rangle$$

there is no U

$$\langle \phi | \psi \rangle \underbrace{\langle 0 | 0 \rangle}_1 = (\langle \phi | \psi \rangle)^2$$

$$\underbrace{\langle \phi | \psi \rangle}_{0 \text{ or } 1} = |\langle \phi | \psi \rangle|^2$$

Suppose there exist a unitary gate U

$$|\psi\rangle|0\rangle \xrightarrow{U} |\psi\rangle|\psi\rangle$$

$$|\phi\rangle|0\rangle \longrightarrow |\phi\rangle|\phi\rangle$$

U must preserve inner product

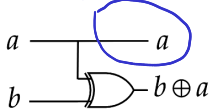
XOR logic gate and Controlled-NOT quantum gate

$$a, b \in \{0, 1\} = \mathbb{Z}_2$$

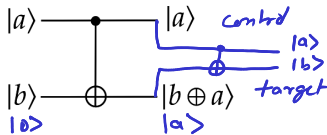
Boolean f^n

$$f: \{0, 1\}^n \rightarrow \{0, 1\}$$

To make it reversible



$$(CNOT)^{-1} = (CNOT)^{\dagger} = CNOT$$



$$|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$|1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Input

a	b	a ⊕ b
0	0	0
0	1	1
1	0	1
1	1	0

Output

module-2 add

$$CNOT = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X \equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Control

target

$$CNOT |a\rangle|b\rangle = |0\rangle\langle 0|a\rangle \otimes |b\rangle + |1\rangle\langle 1|a\rangle \otimes X|b\rangle$$

CNOT

$ a\rangle b\rangle$	$\longleftrightarrow$	$ a\rangle a \oplus b\rangle$
$ 00\rangle$	$\longleftrightarrow$	$ 00\rangle$
$ 01\rangle$	$\longleftrightarrow$	$ 01\rangle$
$ 10\rangle$	$\longleftrightarrow$	$ 11\rangle$
$ 11\rangle$	$\longleftrightarrow$	$ 10\rangle$

Comp. Basis $\mathbb{Z} \otimes \mathbb{Z}$

$$B_Z := \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$|00\rangle$ $|01\rangle$ $|10\rangle$ $|11\rangle$

$(\begin{pmatrix} 1 \\ 0 \end{pmatrix}) \otimes (\begin{pmatrix} 1 \\ 0 \end{pmatrix})$ $(\begin{pmatrix} 0 \\ 1 \end{pmatrix}) \otimes (\begin{pmatrix} 1 \\ 0 \end{pmatrix})$

